

SOUTHWEST WYOMING, WY, USA

WMO: 725744

Lat: 41.595N

Lon: 109.053W

Elev: 6742

StdP: 11.45

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24027

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -4.6	(c) 1.1	(d) -10.3	(e) 4.1	(f) -2.7	(g) -5.3	(h) 5.3	(i) 5.4	(j) 37.0	(k) 26.9	(l) 34.3	(m) 25.9	(n) 8.2	(o) 60	(p) 0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 27.9	(c) 88.3	(d) 55.2	(e) 85.9	(f) 54.4	(g) 83.4	(h) 53.7	(i) 58.7	(j) 78.1	(k) 57.2	(l) 76.7	(m) 55.9	(n) 76.0	(o) 12.6	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
52.8	76.3	62.0	50.3	69.7	61.7	48.0	63.7	61.6	28.3	77.6	27.2	76.4	26.3	75.8	67.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 32.9	(b) 28.7	(c) 26.0	(e) -12.8	(f) 92.5	(g) 5.4	(h) 2.0	(i) -16.7	(j) 93.9	(k) -19.9	(l) 95.1	(m) -22.9	(n) 96.2	(o) -26.9	(p) 97.6		
(a) 32.9	(b) 28.7	(c) 26.0	(e) -13.4	(f) 61.0	(g) 5.3	(h) 2.0	(i) -17.2	(j) 62.4	(k) -20.3	(l) 63.6	(m) -23.3	(n) 64.7	(o) -27.1	(p) 66.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	22.3	24.7	34.0	41.1	50.5	61.3	70.0	67.4	57.6	44.4	31.9	22.0	
	DBStd	18.73	9.82	9.67	9.06	8.46	8.49	7.70	4.58	4.78	8.09	9.29	9.85	10.00	
	HDD50	4118	859	709	498	283	100	11	0	0	30	215	545	868	
	HDD65	7977	1324	1129	961	716	450	153	13	31	235	638	994	1333	
	CDD50	1949	0	0	2	17	117	351	621	539	259	42	1	0	
	CDD65	334	0	0	0	0	2	43	169	106	14	0	0	0	
	CDH74	4548	0	0	0	1	67	743	2107	1347	277	6	0	0	
	CDH80	1383	0	0	0	0	7	199	775	369	33	0	0	0	
Wind	WSAvg	11.2	12.5	12.5	12.3	12.8	11.5	10.8	9.1	8.9	9.5	11.2	11.3	11.8	
Precipitation	PrecAvg	8.2	0.4	0.4	0.6	0.8	1.1	1.0	0.7	0.7	0.8	0.8	0.4	0.4	
	PrecMax	17.8	1.4	1.5	1.9	2.8	4.6	3.4	3.3	2.5	3.2	2.4	1.6	1.5	
	PrecMin	3.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	PrecStd	2.7	0.4	0.4	0.5	0.6	1.0	0.8	0.7	0.6	0.7	0.6	0.4	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.0	49.9	63.5	70.5	80.1	89.4	92.3	89.5	83.6	73.7	60.3	48.2	
		MCWB	35.3	37.6	42.6	46.1	51.3	54.6	57.6	55.9	53.6	48.6	42.0	36.3	
	2%	DB	41.3	44.5	57.4	65.6	75.2	85.3	89.6	86.4	80.3	68.7	55.0	42.5	
		MCWB	32.7	35.0	40.3	43.7	49.2	53.3	55.9	54.3	51.8	46.7	39.8	33.7	
	5%	DB	37.7	40.7	52.8	61.6	71.6	81.9	87.3	84.1	77.4	64.0	50.5	38.4	
		MCWB	31.2	32.9	38.5	42.0	48.0	52.1	55.2	53.7	50.2	44.6	38.2	31.9	
	10%	DB	35.2	37.4	48.0	57.1	67.2	78.5	84.6	81.7	73.8	59.9	46.2	35.5	
		MCWB	29.9	31.0	36.3	40.3	46.2	51.1	54.5	53.4	49.7	43.1	35.8	30.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.0	38.4	44.4	47.4	53.4	57.3	60.8	60.1	57.4	50.7	43.6	38.0
			MCDB	43.7	47.6	60.4	68.2	74.5	81.6	82.4	77.5	75.4	68.1	56.3	46.5
2%		WB	34.0	35.8	41.3	44.7	50.9	55.2	59.2	58.2	54.8	48.5	40.9	34.6	
		MCDB	39.4	43.2	55.1	62.3	70.1	77.8	79.1	76.7	71.2	63.8	52.7	40.7	
5%		WB	32.0	33.5	39.0	42.8	49.2	53.7	57.8	56.8	53.1	46.3	38.8	32.5	
		MCDB	37.1	39.7	51.4	58.9	67.5	75.7	77.5	75.4	69.8	61.4	49.5	37.7	
10%		WB	30.1	31.7	36.9	41.1	47.5	52.3	56.6	55.4	51.5	44.2	36.6	30.6	
		MCDB	34.8	37.0	47.2	55.7	64.9	74.2	76.9	74.4	68.1	58.1	45.4	35.1	
Mean Daily Temperature Range	5% DB	MDBR	15.6	16.5	19.3	22.2	23.9	27.0	27.9	27.0	25.5	22.0	18.3	15.3	
		MCDBR	16.8	18.6	25.3	28.4	29.2	31.0	30.5	30.0	29.5	27.9	23.6	17.6	
		MCWBR	11.4	12.0	12.9	13.1	12.0	12.0	10.9	11.2	12.2	13.2	13.0	11.7	
	5% WB	MCDBR	15.7	17.1	23.8	26.1	26.7	28.3	26.3	25.4	24.7	25.1	21.8	16.4	
		MCWBR	11.2	11.3	12.7	12.7	11.2	11.2	9.4	9.6	10.3	11.9	12.4	11.4	
Clear-Sky Solar Irradiance	taub	0.228	0.235	0.250	0.279	0.297	0.298	0.327	0.321	0.286	0.254	0.234	0.228		
	taud	2.584	2.595	2.576	2.485	2.480	2.495	2.433	2.447	2.543	2.629	2.613	2.587		
	Ebn at Noon	306	318	322	316	310	308	298	298	304	306	301	298		
	Edh at Noon	21	24	28	33	34	34	36	34	29	23	20	19		
All-Sky Solar Radiation	RadAvg	693	980	1398	1790	2087	2423	2286	2010	1679	1178	777	569		
	RadStd	51	81	105	105	139	142	93	69	110	73	68	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page